

Lab Report: Fire Detector Project

1. Project Title

Fire Detector Using IR Receiver and BC547 Transistor

2. Objective

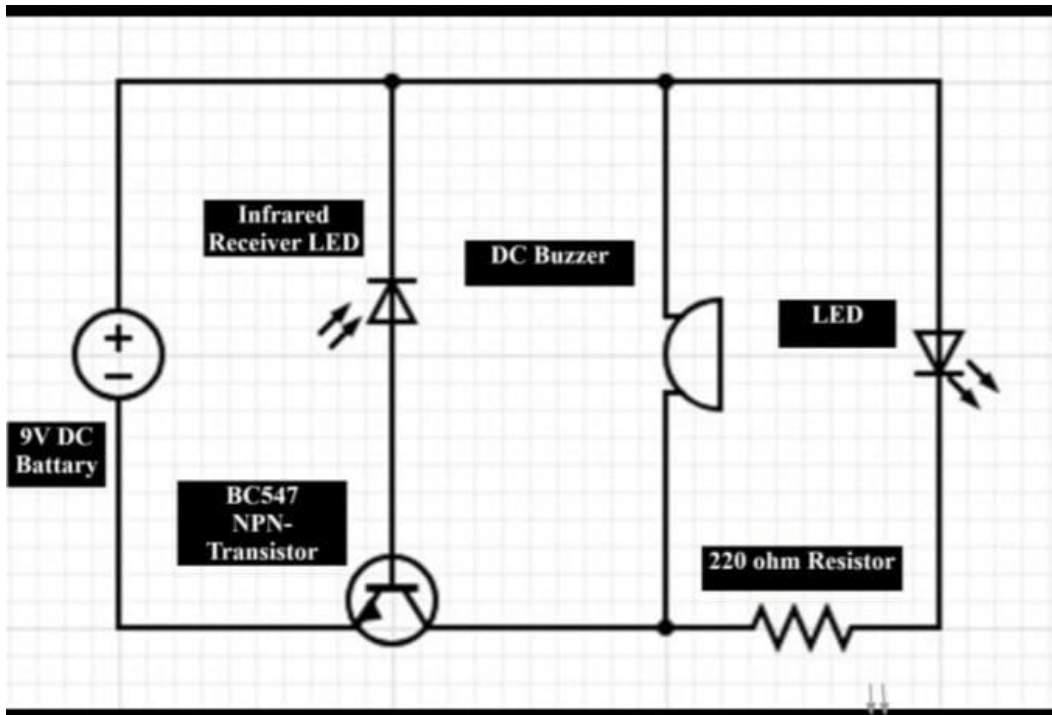
The aim of this project is to design a simple fire detection system that activates a buzzer and LED light when it detects a flame or infrared radiation from a fire source. This low-cost and beginner-friendly circuit uses basic electronic components and demonstrates the principle of IR flame detection.

3. Components Used

Sl. No.	Component	Quantity
1	BC547 NPN Transistor	1
2	IR Receiver LED	1
3	LED Light (Indicator)	1
4	220 Ohm Resistor	1
5	DC Buzzer	1
6	9V Battery	1
7	Breadboard	1
8	Jumper Wires	As required

4. Circuit Diagram

The circuit diagram below represents the fire detection system using IR Receiver, BC547 Transistor, DC Buzzer, and LED.



5. Working Principle

The IR receiver LED is used to detect infrared radiation (which is emitted by fire or flame). When IR radiation is detected:

- It allows a small current to flow into the base of the BC547 transistor.
- This turns on the transistor, allowing a larger current to flow from collector to emitter.
- As a result, the DC buzzer and LED light connected to the collector get activated.
- The system thus gives both a visual (LED) and audible (buzzer) alert indicating fire presence.

6. Construction Steps

1. Place the IR receiver on the breadboard.
2. Connect its output through a 220-ohm resistor to the base of the BC547 transistor.
3. Connect the emitter of the transistor to the ground (-ve terminal of the battery).
4. Connect the LED and buzzer to the collector of the transistor, with their other ends going to the positive terminal of the battery.
5. Ensure the correct polarity for all components, especially the transistor, LED, and buzzer.
6. Power the circuit with a 9V battery.

7. Testing and Observation

Test Condition	Observation
No flame near sensor	LED and buzzer are OFF
Flame (lighter/candle) near IR sensor	LED and buzzer turn ON

8. Conclusion

The fire detector circuit successfully detects the presence of a flame by sensing IR radiation. When triggered, it provides an immediate audio-visual alert, demonstrating a basic fire alarm functionality using simple and low-cost components.

9. Applications

- Home fire detection systems
- Small-scale industrial safety alarms
- Educational electronics projects

10. Limitations

- Limited Detection Range: The IR sensor can only detect flames within a short distance.
 - False Alarms: It may trigger due to other IR sources like sunlight or remote controls.
 - No Smoke Detection: It only detects flames, not smoke or heat buildup.
 - No Remote Alert: Cannot send alerts to phones or remote systems.
- Battery Dependent: System stops working if the battery runs out.

11. Future Improvements

- Adding a relay module to control larger alarms or fans.
- Including an automatic fire suppression system.
- Integrating the circuit with a microcontroller (like Arduino) for remote alerts or data logging.